Second-Order Linear Homogeneous Equations

ANSWER KEY

The auxiliary equation

1 Write the auxiliary (characteristic) equation for $y'' - 5y' + 6y = 0$.
$r^2 - 5r + 6 = 0$.

2 Write and simplify the auxiliary equation for $2y'' + 4y' - 6y = 0$.
$2r^2 + 4r - 6 = 0$, which simplifies to $r^2 + 2r - 3 = 0$.

3 Explain why substituting $y = e^{rx}$ into $ay'' + by' + cy = 0$ produces the auxiliary equation $ar^2 + br + c = 0$.
$y = e^{rx}$ gives $y' = re^{rx}$ and $y'' = r^2e^{rx}$. Substituting: $ar^2e^{rx} + bre^{rx} + ce^{rx} = 0$, i.e., $e^{rx}(ar^2 + br + c) = 0$. Since $e^{rx} \neq 0$ for all x, this forces $ar^2 + br + c = 0$.

Distinct real roots

4 Solve $y'' - 5y' + 6y = 0$.
$r^2 - 5r + 6 = (r-2)(r-3) = 0 \Rightarrow r = 2, 3$. General solution: $y = C_1e^{2x} + C_2e^{3x}$.

5 Solve $y'' + 2y' - 3y = 0$, $y(0) = 1$, $y'(0) = 1$.
$r^2 + 2r - 3 = (r+3)(r-1) = 0 \Rightarrow r = -3, 1$. So $y = C_1e^{-3x} + C_2e^{x}$. From $y(0) = 1$: $C_1 + C_2 = 1$. From $y'(0) = 1$: $-3C_1 + C_2 = 1$. Subtracting gives $4C_1 = 0$, so $C_1 = 0$, $C_2 = 1$. Thus $y = e^x$.

6 Solve $y'' - y' - 6y = 0$.
$r^2 - r - 6 = (r-3)(r+2) = 0 \Rightarrow r = 3, -2$. General solution: $y = C_1e^{3x} + C_2e^{-2x}$.

7 Solve $y'' - 9y = 0$ (note this factors as a difference of squares in r).
$r^2 - 9 = (r-3)(r+3) = 0 \Rightarrow r = \pm 3$. General solution: $y = C_1e^{3x} + C_2e^{-3x}$.

Repeated roots

8 Solve $y'' - 6y' + 9y = 0$.
$r^2 - 6r + 9 = (r-3)^2 = 0 \Rightarrow r = 3$ (repeated). General solution: $y = (C_1 + C_2x)e^{3x}$.

9 Solve $y'' + 4y' + 4y = 0$, $y(0) = 2$, $y'(0) = -1$.
$r^2 + 4r + 4 = (r+2)^2 = 0 \Rightarrow r = -2$. So $y = (C_1 + C_2x)e^{-2x}$. From $y(0) = 2$: $C_1 = 2$. Differentiating, $y' = C_2e^{-2x} - 2(C_1 + C_2x)e^{-2x}$; $y'(0) = C_2 - 2C_1 = -1 \Rightarrow C_2 = 2(2) - 1 = 3$. Thus $y = (2 + 3x)e^{-2x}$.

10 Explain why, for a repeated root r, $y_2 = xe^{rx}$ (not just a second copy of e^{rx}) is needed to build the general solution.

$y_1 = e^{rx}$ and a constant multiple of it, ce^{rx}, are linearly dependent — they don't give two independent solutions needed for a second-order equation's two-dimensional solution space. Direct substitution confirms $y_2 = xe^{rx}$ also solves the ODE when r is a repeated root, and y_1, y_2 are linearly independent (their Wronskian is $e^{2rx} \neq 0$), so together they span the full solution space.

Complex roots

11 Solve $y'' + 4y = 0$.

$r^2 + 4 = 0 \Rightarrow r = \pm 2i$. General solution: $y = C_1 \cos 2x + C_2 \sin 2x$.

12 Solve $y'' - 2y' + 5y = 0$.

$r^2 - 2r + 5 = 0 \Rightarrow r = \dfrac{2 \pm \sqrt{4 - 20}}{2} = 1 \pm 2i$. General solution: $y = e^x(C_1 \cos 2x + C_2 \sin 2x)$.

13 Solve $y'' + 9y = 0$, $y(0) = 2$, $y'(0) = 3$.

$r^2 + 9 = 0 \Rightarrow r = \pm 3i$. So $y = C_1 \cos 3x + C_2 \sin 3x$. $y(0) = C_1 = 2$. $y' = -3C_1 \sin 3x + 3C_2 \cos 3x$; $y'(0) = 3C_2 = 3 \Rightarrow C_2 = 1$. Thus $y = 2\cos 3x + \sin 3x$.

14 Solve $y'' + 2y' + 2y = 0$.

$r^2 + 2r + 2 = 0 \Rightarrow r = \dfrac{-2 \pm \sqrt{4 - 8}}{2} = -1 \pm i$. General solution: $y = e^{-x}(C_1 \cos x + C_2 \sin x)$.

Classifying without fully solving

15 Without solving, state whether $y'' + 6y' + 5y = 0$ has real, repeated, or complex roots, using the discriminant $b^2 - 4ac$ of the auxiliary equation.

Discriminant $= 36 - 20 = 16 > 0$ and a perfect square, so the roots are real and distinct ($r = -1, -5$).

16 Without solving, classify the roots of $y'' + 10y' + 25y = 0$.

Discriminant $= 100 - 100 = 0$, so the auxiliary equation has one repeated real root ($r = -5$).

17 Without solving, classify the roots of $y'' + 2y' + 10y = 0$.

Discriminant $= 4 - 40 = -36 < 0$, so the roots are complex conjugates ($r = -1 \pm 3i$).

Application: a damped spring

18 A mass–spring–damper system satisfies $y'' + 4y' + 5y = 0$. Find the general solution and classify the damping (over/critically/under-damped).

$r^2 + 4r + 5 = 0$, discriminant $= 16 - 20 = -4 < 0$, so $r = -2 \pm i$. General solution: $y = e^{-2x}(C_1 \cos x + C_2 \sin x)$ — this is under-damped (complex roots with negative real part give oscillatory decay).

19 Solve $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.

$r^2 + 6r + 9 = (r+3)^2 = 0 \Rightarrow r = -3$ (repeated) — critically damped. So $y = (C_1 + C_2 x)e^{-3x}$. From $y(0) = 1$: $C_1 = 1$. $y' = C_2 e^{-3x} - 3(C_1 + C_2 x)e^{-3x}$; $y'(0) = C_2 - 3C_1 = 0 \Rightarrow C_2 = 3$. Thus $y = (1 + 3x)e^{-3x}$.

20 Solve $y'' + 5y' + 6y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.

$r^2 + 5r + 6 = (r+2)(r+3) = 0 \Rightarrow r = -2, -3$ (real, distinct, both negative) — over-damped. So $y = C_1 e^{-2x} + C_2 e^{-3x}$. $y(0) = C_1 + C_2 = 1$. $y' = -2C_1 e^{-2x} - 3C_2 e^{-3x}$; $y'(0) = -2C_1 - 3C_2 = 0 \Rightarrow C_1 = -\frac{3}{2}C_2$. Substituting: $-\frac{3}{2}C_2 + C_2 = 1 \Rightarrow -\frac{1}{2}C_2 = 1 \Rightarrow C_2 = -2$, $C_1 = 3$. Thus $y = 3e^{-2x} - 2e^{-3x}$.

21 For the general damped spring equation $my'' + cy' + ky = 0$ ($m, c, k > 0$), explain in one sentence why the real part of every root of the auxiliary equation is always negative (so every solution decays).

By the quadratic formula the roots are $r = \dfrac{-c \pm \sqrt{c^2 - 4mk}}{2m}$: when the roots are real, $\sqrt{c^2 - 4mk} < c$ (since $mk > 0$), so both roots stay negative after adding $-c$ and dividing by $2m > 0$; when complex, the real part is exactly $-\dfrac{c}{2m} < 0$ since $c, m > 0$ — either way the exponential factor $e^{-(c/2m)x}$ (or the two negative real exponentials) forces every solution toward 0 as $x \to \infty$.

Verifying general solutions and the superposition principle

22 Verify by direct substitution that $y = C_1 e^{2x} + C_2 e^{3x}$ solves $y'' - 5y' + 6y = 0$ for any constants C_1, C_2.

$y' = 2C_1 e^{2x} + 3C_2 e^{3x}$, $y'' = 4C_1 e^{2x} + 9C_2 e^{3x}$. Then $y'' - 5y' + 6y = (4 - 10 + 6)C_1 e^{2x} + (9 - 15 + 6)C_2 e^{3x} = 0 + 0 = 0$ for any C_1, C_2.

23 State the superposition principle for a linear homogeneous ODE, and explain why it guarantees that $y = C_1 y_1 + C_2 y_2$ solves $ay'' + by' + cy = 0$ whenever y_1, y_2 each do.

If y_1, y_2 both solve $ay'' + by' + cy = 0$, then so does any linear combination $C_1 y_1 + C_2 y_2$, because the operator $y \mapsto ay'' + by' + cy$ is linear: $a(C_1 y_1 + C_2 y_2)'' + b(\cdots)' + c(\cdots) = C_1(ay_1'' + by_1' + cy_1) + C_2(ay_2'' + by_2' + cy_2) = C_1(0) + C_2(0) = 0$.

24 Find the auxiliary equation's roots for $y'' - 4y = 0$, then verify that both $y = e^{2x}$ and $y = e^{-2x}$ individually satisfy the ODE.

$r^2 - 4 = 0 \Rightarrow r = \pm 2$. For $y = e^{2x}$: $y'' = 4e^{2x}$, so $y'' - 4y = 4e^{2x} - 4e^{2x} = 0$. For $y = e^{-2x}$: $y'' = 4e^{-2x}$, so $y'' - 4y = 4e^{-2x} - 4e^{-2x} = 0$. Both check out.

Boundary value problems

25 Solve the boundary value problem $y'' + y = 0$, $y(0) = 0$, $y(\pi/2) = 3$.

$r^2 + 1 = 0 \Rightarrow r = \pm i$, so $y = C_1 \cos x + C_2 \sin x$. $y(0) = C_1 = 0$. $y(\pi/2) = C_2 \sin(\pi/2) = C_2 = 3$. Thus $y = 3\sin x$.

26 Solve $y'' - y = 0$, $y(0) = 1$, $y(1) = e$.

$r^2 - 1 = 0 \Rightarrow r = \pm 1$, so $y = C_1 e^x + C_2 e^{-x}$. $y(0) = C_1 + C_2 = 1$. $y(1) = C_1 e + C_2 e^{-1} = e$. From the first equation, $C_2 = 1 - C_1$; substituting: $C_1 e + (1 - C_1) e^{-1} = e \Rightarrow C_1(e - e^{-1}) = e - e^{-1} \Rightarrow C_1 = 1$, so $C_2 = 0$. Thus $y = e^x$.

27 Explain why the boundary value problem $y'' + 4y = 0$, $y(0) = 0$, $y(\pi) = 1$ has NO solution, while $y(0) = 0$, $y(\pi/4) = 1$ does.

General solution: $y = C_1 \cos 2x + C_2 \sin 2x$. $y(0) = C_1 = 0$, so $y = C_2 \sin 2x$. At $x = \pi$: $\sin 2\pi = 0$, so $y(\pi) = 0$ for every C_2 — the condition $y(\pi) = 1$ can never be met. At $x = \pi/4$: $\sin(\pi/2) = 1 \neq 0$, so $C_2 = 1$ solves $y(\pi/4) = 1$ uniquely.